
Chapter 9: Area and Its Boundary (Mensuration)

Worksheet 9: Surface Area and Volume of Solids

Learning Objective: Students will calculate surface area (total and lateral) and volume of cubes, cuboids, cylinders, and cones; solve problems involving composite solids; and apply mensuration to real-life contexts.

Worked Example

Find the total surface area and volume of a cylinder with radius 7 cm and height 10 cm. (Use $\pi = \frac{22}{7}$)

Solution:

$$\text{TSA} = 2\pi r(r + h) = 2 \times \frac{22}{7} \times 7 \times 17 = 748 \text{ sq. cm}$$

$$\text{Volume} = \pi r^2 h = \frac{22}{7} \times 49 \times 10 = 1540 \text{ cubic cm}$$

Questions 1–5: Easy / Recall

Q1. Find the volume of a cube with edge 6 cm.

Q2. Find the TSA of a cuboid: $l = 8$ cm, $b = 5$ cm, $h = 3$ cm.

Q3. Find the volume of a cylinder: $r = 3.5$ cm, $h = 10$ cm. (Use $\pi = \frac{22}{7}$)

Q4. Find the lateral (curved) surface area of a cylinder: $r = 7$ cm, $h = 15$ cm.

Q5. A cube has TSA = 294 sq. cm. Find its edge.

Questions 6–10: Basic Application

Q6. A cuboidal box is $12\text{ cm} \times 8\text{ cm} \times 5\text{ cm}$. Find the area of cardboard needed to make it (TSA).

Q7. Find the volume of a cylindrical tank: diameter 14 m, height 5 m.

Q8. How many bricks (each $25\text{ cm} \times 12.5\text{ cm} \times 7.5\text{ cm}$) are needed to build a wall $10\text{ m} \times 3\text{ m} \times 0.5\text{ m}$?

Q9. The radius of a cylinder is doubled and height halved. How does the volume change?

Q10. A cylindrical well has diameter 2 m and depth 14 m. Find the volume of earth dug out.

Questions 11–15: Practice and Skill Building

Q11. Find the volume of water in a cylindrical tank (diameter 3.5 m, height 4 m) that is $\frac{2}{3}$ full.

Q12. A cuboid has volume 840 cubic cm. Its length is 10 cm and breadth is 7 cm. Find its height and TSA.

Q13. A cube of metal with edge 8 cm is melted and recast into 64 smaller cubes. Find the edge of each small cube.

Q14. Find the height of a cylinder whose volume equals a cube of edge 11 cm. The cylinder's radius is 7 cm. (Use $\pi = \frac{22}{7}$)

Q15. A road roller is a cylinder of radius 0.7 m and length 2 m. Find the area it flattens in one revolution.

Questions 16–18: Higher Order Thinking

Q16. A cylindrical pipe of outer diameter 12 cm and inner diameter 10 cm is 1 m long. Find the volume of metal in the pipe.

Q17. A hall is $20 \text{ m} \times 15 \text{ m} \times 5 \text{ m}$. It has 3 windows (each $1.5 \text{ m} \times 1.2 \text{ m}$) and 2 doors (each $2 \text{ m} \times 1 \text{ m}$). Find the area of four walls excluding windows and doors. If painting costs Rs. 80 per sq. m, find the cost.

Q18. A toy is shaped like a cylinder with a cone on top. Cylinder: radius 3 cm, height 8 cm. Cone: same base radius, height 4 cm. Find the total surface area and volume of the toy. (Use $\pi = \frac{22}{7}$; cone LSA = πrl where l = slant height.)

Questions 19–20: Real-Life Applications

Q19. A water tanker (cylinder) has radius 1 m and length 7 m. It supplies water to a colony where each family needs 200 litres per day. How many families can the tanker serve in one trip? (1 cubic m = 1000 litres)

- Q20.** A farmer digs a cylindrical well of diameter 3 m and depth 14 m. The earth dug out is spread evenly on a $20\text{ m} \times 15\text{ m}$ rectangular plot. Find the height of the earth on the plot. (Use $\pi = \frac{22}{7}$)

Reflection Question: Why does a cylinder hold more volume than a cube of the same surface area? Explore with an example.